

- 2 A simple test can be used to assess the quantity of sugar in liquids, such as food extracts and urine.

(a) (i) Describe how a urine sample can be tested for glucose.

(2)

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(ii) Complete the table of results from tests on standard glucose solutions.

(2)

Appearance	Glucose concentration / g dm ⁻³	Glucose quantity
Blue	0.0	None
Green with no visible precipitate	0.1 to 0.4	Trace
Green with precipitate	0.5 to 10.0	Very low
Yellow with precipitate		Low
Orange with precipitate	15.1 to 20.0	Moderate
	>20.0	High



- (b) People with type 2 diabetes can have a raised blood glucose concentration.

The concentration of glucose in the urine can be used to identify people with raised blood glucose levels.

Blood glucose concentrations can be estimated using test strips.

A study measured the urine glucose concentrations and the blood glucose concentrations of a group of people with type 2 diabetes.

The table shows the results of this study.

Mean urine glucose concentration / a.u.	Mean blood glucose concentration / a.u.	Standard deviation of blood glucose concentration / a.u.
0	150	± 140
2	225	± 170
4	280	± 170
6	315	± 185
8	370	± 140
10	415	± 165

- (i) Describe how the measurements for blood glucose concentration taken in this study were processed to give the data in the table.

(2)

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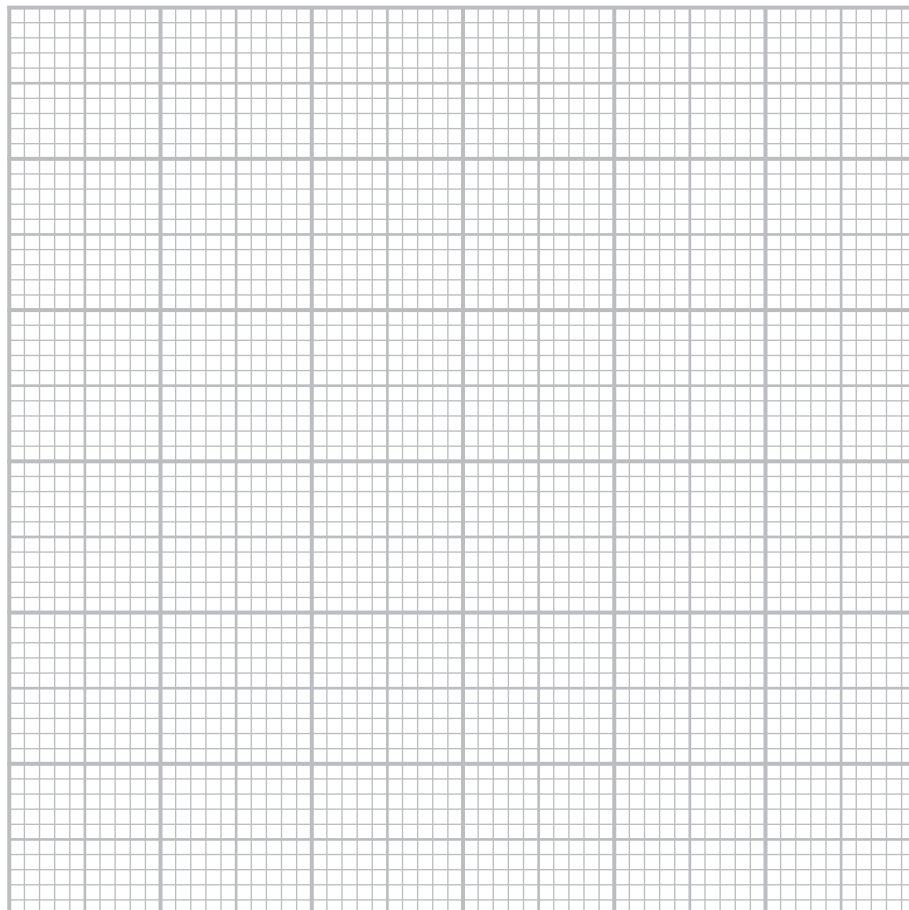


- (ii) Plot a suitable graph to show the relationship between mean urine glucose concentration and mean blood glucose concentration.

Plot the standard deviation for the mean blood glucose concentration at the mean urine glucose concentration of 2 a.u.

Draw a line of best fit.

(5)



- (iii) Comment on the conclusion that urine glucose concentration is a good indicator of the blood glucose level.

Use the information in the table and in your graph to support your answer.

(3)

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- (iv) Calculate the blood glucose concentration at a urine glucose concentration of 12 a.u.

The equation for the line of best fit is $y = 29.3x + 154.1$.

Give your answer to 3 significant figures.

(3)

Answer a.u.

(Total for Question 2 = 17 marks)

